

TRAJECTORY

AI-Based Network Attack Forecasting from Network Traffic Data

An offline, explainable temporal cyber-defence system that forecasts likely attack progression before compromise is complete.



PROBLEM STATEMENT ID

SIH26153

PROBLEM STATEMENT TITLE

AI based Network Attack Forecasting from Network Traffic Data

ORGANIZATION

National Technical Research Organisation (NTRO)

THEME

Blockchain & Cybersecurity

PS CATEGORY

Software

TEAM ID

[ADD OFFICIAL TEAM ID]

TEAM NAME

[ADD OFFICIAL TEAM NAME]

INSTITUTE

[ADD OFFICIAL INSTITUTE NAME]

Solution Overview & Prototype

From static alerts to predictive attack intelligence

Trajectory ingests flow CSV and PCAP-derived packet data, converts them into ordered network states, and forecasts likely attack progression over K future windows: risk, predicted stage, affected assets, confidence and supporting evidence in an offline analyst dashboard (Web/desktop).

SOC Analyst

- Load CSV flows / PCAP captures
- Risk timeline over time windows
- Next likely stage + affected assets
- Evidence behind every forecast
- Predicted vs actual replay

Incident Responder

- Prioritize hosts/segments at risk
- Confidence + insufficient-evidence flags
- Decision support, not auto-blocking
- Export reproducible forecast record

Security Lead / Evaluator

- Temporal model vs logistic baseline
- Precision, recall, F1, FPR, calibration
- Forecast lead time
- Deterministic, seed-fixed replay
- Fully local; no cloud AI

Project Status

- Phase: ingestion + temporal-data foundation done
- CSV & PCAP ingestion, flow + packet features
- Timestamped overlapping state windows
- Transition targets, leakage-safe splits
- 15 tests passing; baseline model is next

Why We Stand Out

- Predictive, not retrospective
- K-step future-state simulation
- Temporal context instead of per-flow labels
- Evidence + assets with every forecast
- Offline-first; honest baseline comparison

PROTOTYPE VISUALS

1. Data Workspace

CSV/PCAP input | event count, time range | feature-coverage indicators

2. Forecast Dashboard

Risk over K windows | Current / Forecast / Actual replay | stage + confidence

3. Evidence View

Driving features | Server-03, failed-auth burst, destination diversity | source events

ILLUSTRATIVE OUTPUT

Current state: Suspicious reconnaissance
Predicted stage: Lateral movement (+3 windows)
Likely asset: Server-03 Confidence: 0.72
Evidence: failed-auth burst, new internal connections, rising destination diversity

Backend Architecture & Technical Approach

Local, reproducible, evidence-first forecasting pipeline

INPUT	Flow CSV PCAP captures JSONL/syslog tail Scapy live capture
DATA	Unified events -> timestamped windows -> state store labels, transition targets, split manifests
MODEL	Logistic Regression baseline (current window) GRU/LSTM temporal model + K-step rollout
OUTPUT	Risk timeline attack stage assets evidence confidence + warnings
API	FastAPI REST: /v1/forecast, /v1/detect, /v1/alerts — same checksummed artifacts as the UI
UI	Offline analyst dashboard (Streamlit + Plotly)

Security & Privacy

- Local-first; no external AI service
- API keys hashed (SHA-256) + RBAC
- Anonymized identifiers in demos
- No secrets/captures committed
- Missing input -> error, never a fabricated forecast

Authentication Flow

- POST /admin/keys -> sent_... key shown once
- Only SHA-256 hashes stored
- X-API-Key header on every request
- 401 unauthenticated · 403 wrong role
- Revocation wins; every request audit-logged

ROLE MATRIX (API)

Role	API access
viewer	read-only
analyst	forecast · detect
engineer	+ list keys
admin	full control

TECHNICAL APPROACH FLOW



Security lead configures scenario & evaluation -> Analyst loads telemetry & inspects forecast -> Responder validates evidence & acts

AI COMPONENTS

Component	Technique	Function
Static baseline	Logistic Regression	Fair reference from the current window
Temporal forecaster	GRU / LSTM (proposed)	Learn dependencies across ordered states
Future simulation	Recursive K-step rollout	Likely future states, risk trajectory
Stage mapping	Documented rules / classifier	MITRE ATT&CK-oriented stage label
Explanation	Attribution + event retrieval	Driving features and traffic evidence
Calibration	Reliability analysis	Useful confidence values

Implemented: FastAPI REST + hashed API keys with role matrix + audit log. No Kafka/cloud gateway/microservices claimed; prototype runs on local files and artifacts.

Feasibility, Viability & Challenges

Feasible with open data, local compute and controlled deployment

Technical

- Public flow & PCAP datasets available
- Open-source parsers + ML frameworks
- GRU/LSTM practical on dev hardware
- Scenario-safe splits, baseline defined

Operational

- Runs beside IDS / SIEM / EDR
- Analyst owns response decisions
- Deterministic replay for demos & audit
- Local processing for sensitive sites

Economic

- Open-source stack, low prototype cost
- Software-only MVP, no custom hardware
- Augments existing tools
- Pilot-first phased adoption

Regulatory & Privacy

- Synthetic / public / anonymized demo data
- Licences & access restrictions recorded
- Captures and model artifacts stay local
- Human approval; no autonomous enforcement

Market growth (illustrative shape)



CAGR: [ADD VERIFIED VALUE]% Source: [ADD REPORT PUBLISHER, YEAR, URL]
Chart shape is illustrative until a verified market source is inserted, state geography and product scope with the figure.

CHALLENGES -> TECHNICAL RESPONSES

Challenge	Technical response
Rare, imbalanced attacks	Class weighting, per-class metrics, PR analysis
Temporal leakage	Scenario / campaign / time-held-out evaluation
Dataset shift	Secondary-dataset tests, per-site recalibration
Missing packet features	Coverage report + reduced-context warning
Recursive forecast drift	Short K horizon, calibration, uncertainty display
Ambiguous stage labels	Documented mapping; insufficient-evidence state
False-positive fatigue	Baseline comparison, threshold tuning, FPR reporting
Black-box predictions	Attribution, source-event evidence, asset context
Sensitive telemetry	Local processing, anonymization, restricted artifacts
Enterprise integration	File-based MVP first; adapters/APIs later

Viability: feasible as an offline prototype on public datasets and open-source tools. Production use needs environment-specific validation, integration, calibration, access control and security review.

Impacts, Benefits & Stakeholder Scenario

Earlier, evidence-backed decisions can reduce attack impact

Economic

- Analyst time on likely paths
- Less manual correlation
- Earlier intervention, lower downtime cost
- No stack replacement

Social

- Resilient public services & CII
- Clearer incident context
- Humans keep the decision
- Accountable AI use

Environmental

- Runs on ordinary hardware
- Reuses existing telemetry
- No mandatory cloud transfer
- Software-only, no devices

Operational

- Forecast before next stage
- Stage + asset + horizon + evidence
- Repeatable replay for audit
- Layer on existing controls

SUSTAINABLE DEVELOPMENT GOALS

SDG 9

Industry, Innovation & Infrastructure
Resilience of digital and critical infrastructure via predictive security analytics

SDG 16

Peace, Justice & Strong Institutions
Safer digital public systems; accountable, evidence-backed security operations

SAMPLE SCENARIO: RECONNAISSANCE -> LATERAL MOVEMENT



Success criteria: forecast appears before the target stage is observable; evidence names real patterns and entities; predicted vs actual are visually distinct; replay is reproducible from the same configuration.

Use official icons only where template/UN guidance permits.

QUANTITATIVE TARGET IMPACT

Forecast lead time = $t(\text{target stage observable}) - t(\text{warning threshold crossed})$
Time saved / incident = $\text{mean investigation time (current)} - \text{mean time with evidence panel}$
Analyst-hours recovered / year = $\text{valid incidents per year} \times \text{time saved per incident}$

[ADD X] min median forecast lead time
[ADD Y]% investigation-time reduction (controlled user test)
[ADD Z] analyst-hours recovered / year | basis: [ADD DATASET / SCENARIOS / METHOD]

Research, Market Sizing & Business Model

Research-backed; scalable from offline pilot to enterprise integration

Official & Domain

- SIH 2026 PS: sih.gov.in/sih2026PS
- MITRE ATT&CK: attack.mitre.org | CAPEC: capec.mitre.org
- NVD: nvd.nist.gov | NCIIPC: nciipc.gov.in

Datasets

- CIC-IDS2017/2018, UNSW-NB15, CTU-13, CICIoT2023, LANL auth
- URLs: [ADD VERIFIED DATASET URLS]
- Only with documented licence, access, checksums

AI / ML Research

- GRU/LSTM sequence forecasting; recursive multi-step + uncertainty
- Calibration; feature attribution / XAI; scenario-held-out IDS evaluation
- Papers: [ADD 2-4 AUTHOR, TITLE, VENUE, YEAR, DOI]

Software

- PyTorch | scikit-learn | Scapy | NetworkX | Plotly | Streamlit

MARKET SIZING (BOTTOM-UP)

Tier	Definition	Estimate
TAM	Orgs with network-security teams in target geography x annual value	[ADD N1 x A = TAM]
SAM	Reachable with offline product + supported integrations	[ADD N2 x A = SAM]
SOM	Realistic customers in first 3 years	[ADD N3 x A = SOM]

State: geography & sectors, org-count source, scope of annual value, adoption rationale, tax/support inclusion.

REVENUE & HARDWARE

Component	Unit	Pricing basis
Platform licence	Per org / env / year	[ADD VALIDATED PRICE]
Deployment & integration	One-time per env	Data sources + SIEM effort
Model calibration	Per env / telemetry change	Data prep + threshold tuning
Support & updates	Annual	% of licence or tier
Hardware	Not required for MVP	Customer workstation/server

FIRST-YEAR REVENUE (ESTIMATE)

Licences [ADD L] x Rs[licence] = Rs[R1]
Deploys [ADD D] x Rs[deploy] = Rs[R2]
Support [ADD S] x Rs[support] = Rs[R3]
Training [ADD T] x Rs[training] = Rs[R4]

Total (business estimate) = Rs[ADD TOTAL]

PROOF DOCUMENTS

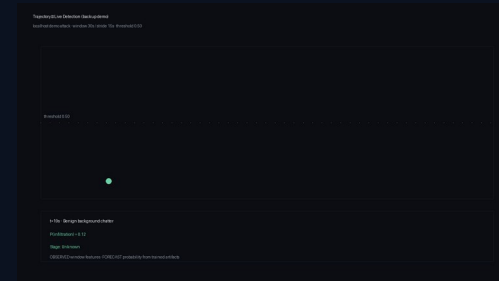
- Source repo: [ADD GITHUB URL]
- Architecture: ARCHITECTURE.md
- Status: IMPLEMENTATION_STATUS.md
- Evaluation report: [ADD LINK AFTER BENCHMARKS]
- Demo video: [ADD FINAL VIDEO LINK]
- Consolidated research: [ADD FOLDER LINK]
- Reproducibility: README.md + uv.lock

Live Demo — Watch the AI Detect an Attack

Real-time detection on a localhost attack simulation; deterministic and rehearsed

Stage Storyline (measured in rehearsal)

- 0:00 — Train the models live (~3 seconds, deterministic seed)
- 0:10 — Press Start: benign chatter begins
- 0:19 — Quiet window: $P(\text{infiltration}) = 0.12$, no false alarms
- 0:34 — ALERT: scan burst detected at $P = 0.99$
 - Stage: Initial Access — MITRE ATT&CK TA0001
- 0:49 — Failed logins: $P = 1.00$, evidence panel grows
- 1:04 — ESCALATION: Lateral Movement — MITRE TA0008
- Same trained artifacts as the offline benchmark
- Sources: demo attack, CIC-IDS2017 replay, live capture



t=19s — benign, $P=0.12$



t=34s — ALERT, $P=0.99$



t=64s — Lateral Movement